EXPERIMENT 8

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AIM: Configuring a Cryptocurrency Wallet using Metamask and integrating it with a DApp

THEORY:

Cryptocurrency Wallet - A wallet provides an interface for managing blockchain accounts and approving transactions. It holds the user's keys and lets the user decide which transactions are signed

and
sent.

MetaMask A browser wallet that connects a decentralized application (DApp) to an Ethereum-compatible network and exposes the connected account to the application. The DApp can request permission to view the account and to send transaction requests, while MetaMask keeps control of approval, so funds cannot be moved without the user's consent.

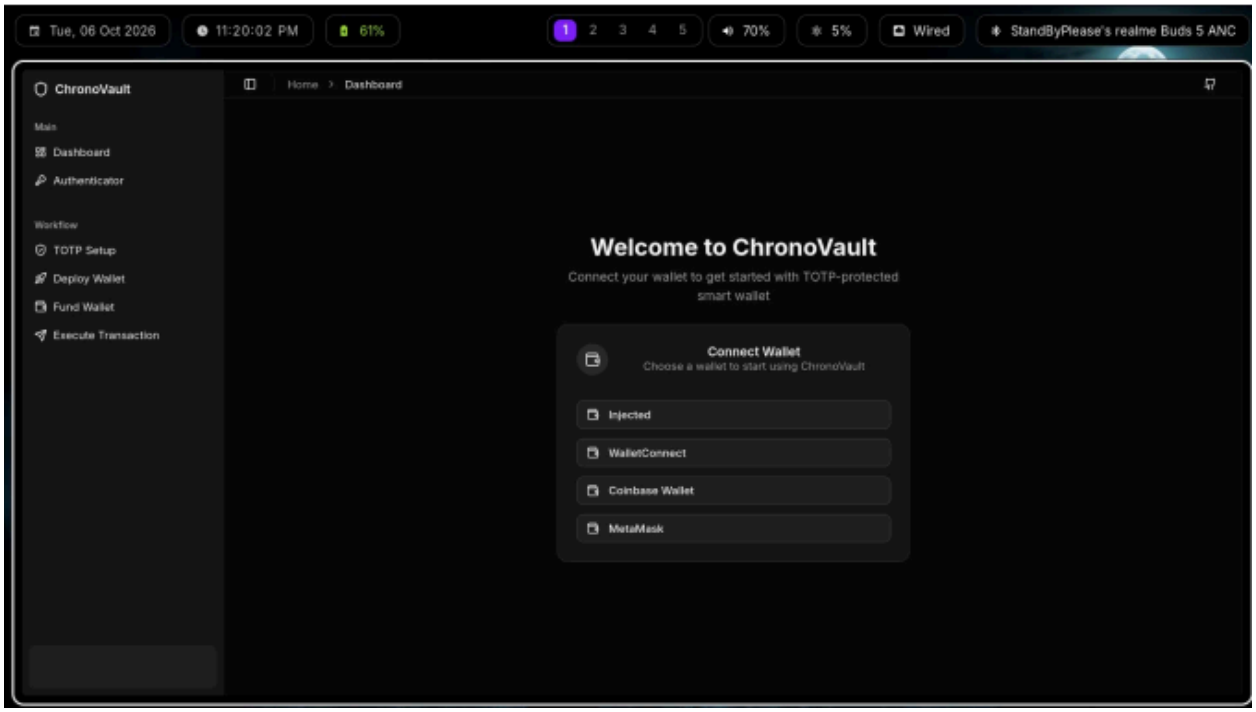
ChronoVault DApp - In this experiment the Chrono Vault DApp uses MetaMask to connect an account. It offers workflow options such as TOTP setup, wallet deployment, funding and transaction execution. The screenshots show the connection request, the network permission confirmation, the connected-wallet dashboard, the account details form and a deployment request on the Sepolia

CODE AND OUTPUT :

Step
1
Open the ChronoVault DApp and view the wallet options

https://chronovault-ethonline.vercel.app

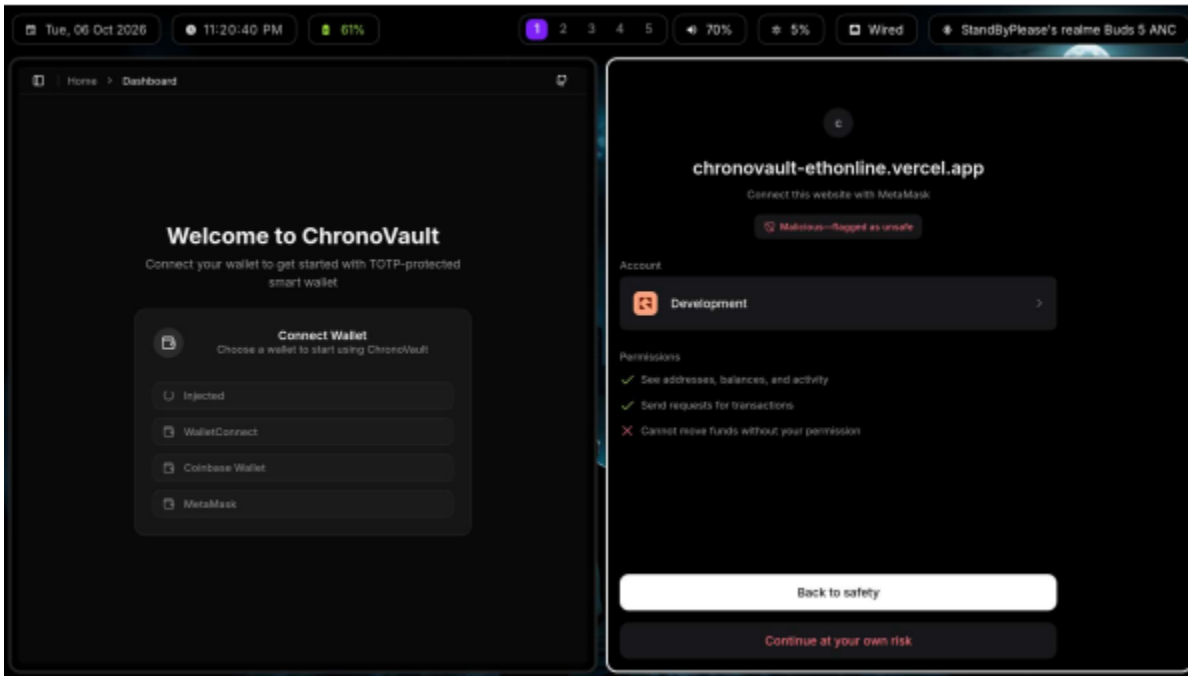
The welcome screen offers Injected, WalletConnect, Coinbase Wallet and MetaMask as connection options.



Step 2

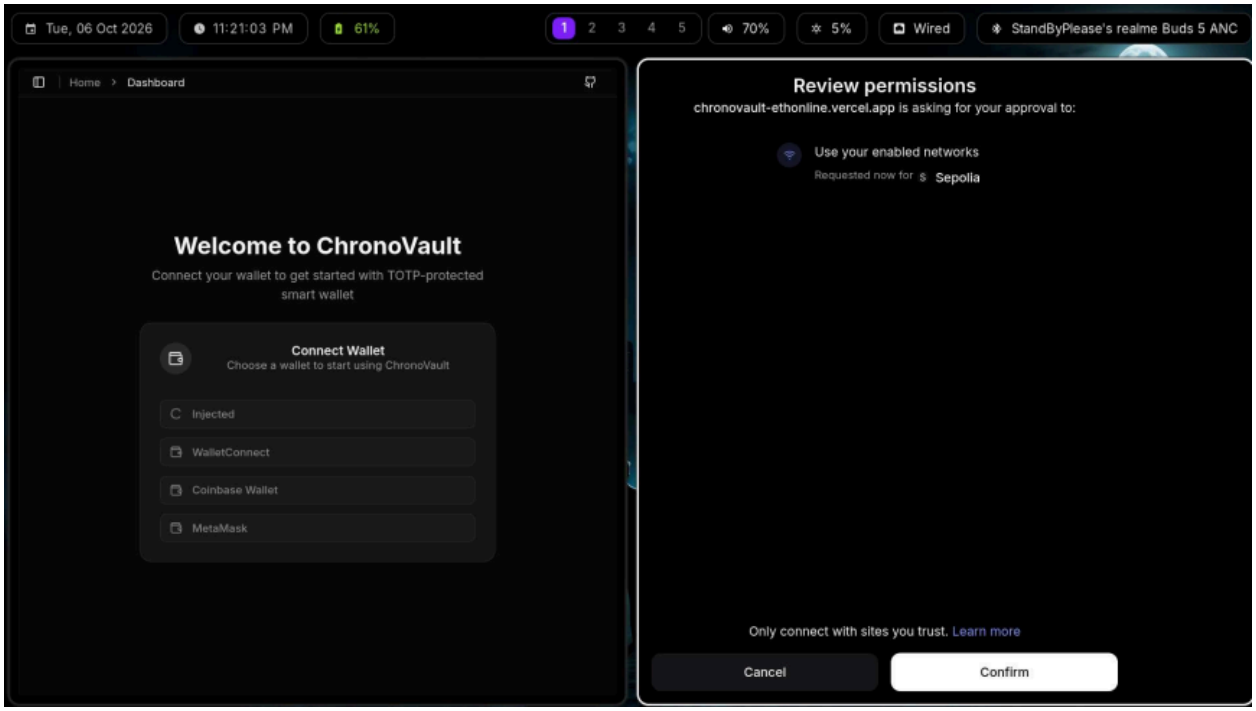
Select MetaMask and review the connection request

MetaMask shows the connection request for chronovault-ethonline.vercel.app with the selected account. The permissions listed are to see addresses, balances and activity and to send requests for transactions, and it cannot move funds without permission.



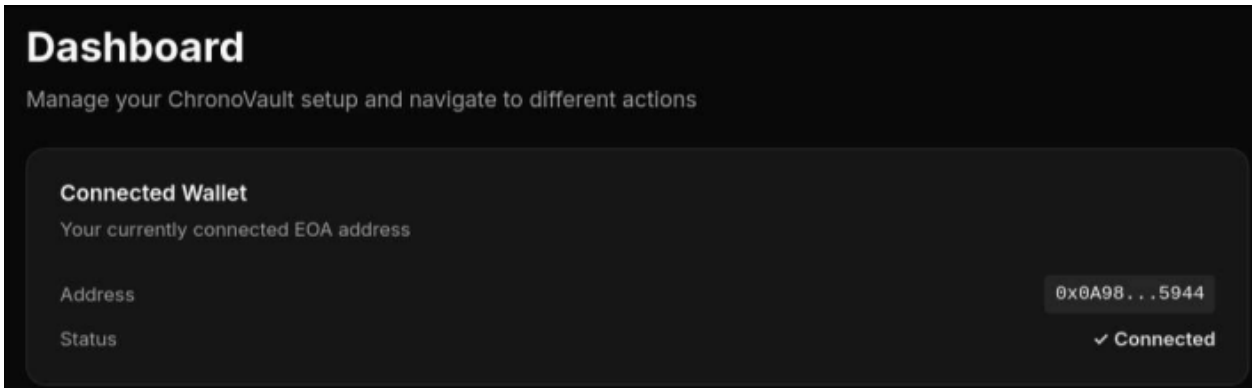
Step 3 Review the network permission

The site asks to use the enabled networks, and the requested network is Sepolia. The connection is approved with Confirm.



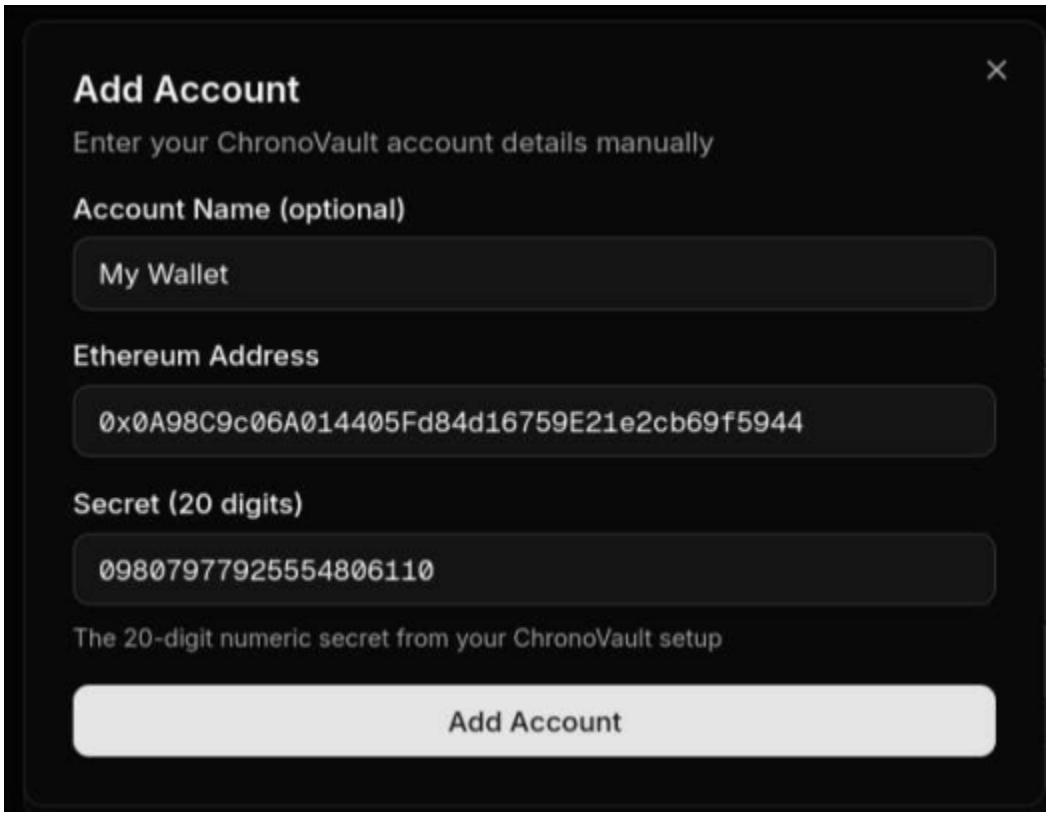
Step 4-
Verify the connected wallet on the dashboard

The dashboard shows the connected EOA address and the status Connected.



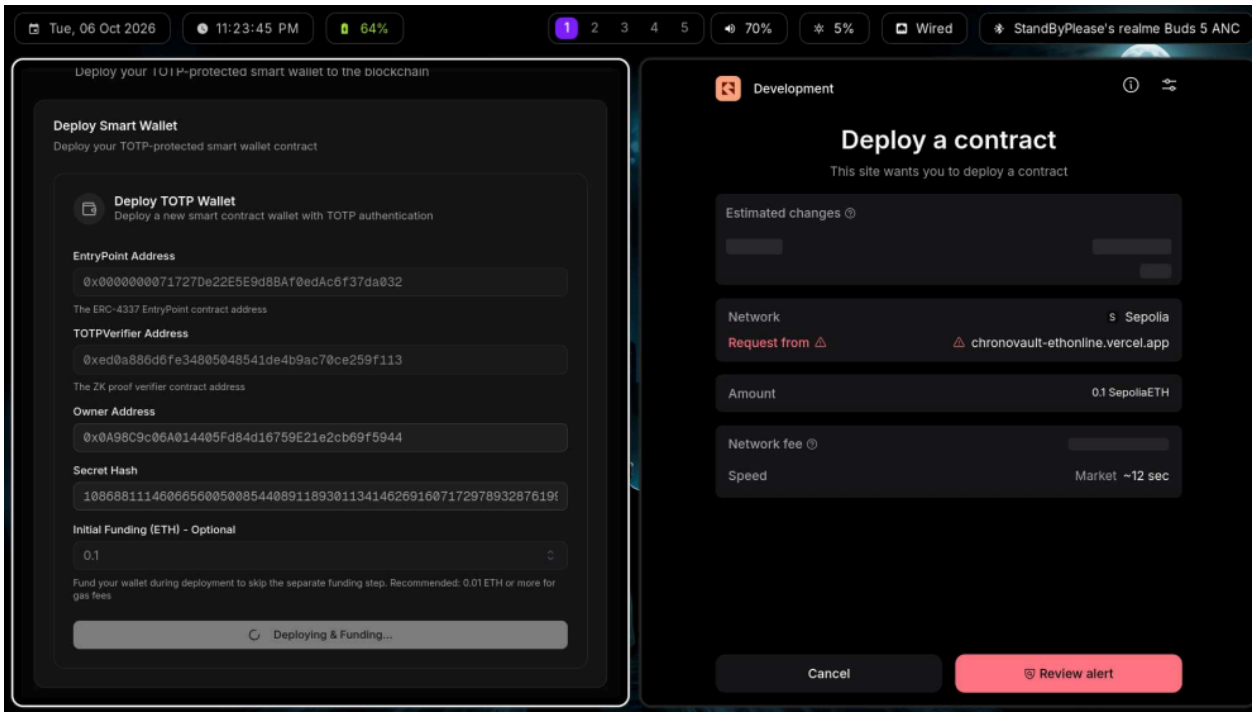
Step 5 - Enter the Chrono Vault account details

In the Add Account form, the account name, Ethereum address and the 20-digit secret from the Chrono Vault setup are entered.



Step 6- Review the smart wallet deployment request

The Deploy TOTP Wallet workflow lists the EntryPoint, TOTPVerifier and owner addresses, the secret hash and the 0.1 ETH initial funding. MetaMask then shows a "Deploy a contract" request on Sepolia for review.



CONCLUSION :

The **MetaMask** wallet was connected to the **ChronoVault DApp** and the requested permissions were reviewed for the **Sepolia** network. The dashboard displayed the connected wallet address and status, the account configuration form was reached successfully, and the deployment workflow presented a smart-wallet deployment request through MetaMask. This demonstrated the **integration of a browser cryptocurrency wallet with a decentralized application**.